

Kinematic cuts buy pre-selection $S/\sqrt{B}$ — they do not buy control-region purity

The MVA defines **purier** control regions *without* any kinematic cut, and already applies those cuts internally when it picks the signal region.

The real lever on SR yield is the **charm-tag working point**.

01 • The $t\bar{t}$ control region is purer when the MVA defines it

Purity = fraction of events whose *true* process is $t\bar{t}$. Pooled MC, 4,046,127 events, raw/unweighted.

T $\bar{T}$ CR DEFINITION	N	T $\bar{T}$ PURITY	SIGNAL CONTAM.	NON-T $\bar{T}$ BKG
argmax = $t\bar{t}$, no kinematic cuts	1,565,461	87.86%	0.0004%	12.14%
$m_{T_{l2}} > 30$ & $m_{T_{ll}} \leq 60$ (<i>classic top CR</i>)	565,368	78.10%	0.0142%	21.89%
$m_{ll} > 72$	2,375,781	83.04%	0.0009%	16.96%
$m_{ll} > 72$ & $m_{T_{l2}} > 30$ & $m_{T_{ll}} > 60$	1,614,248	83.12%	0.0004%	16.88%

The MVA CR wins on both axes at once — purest (87.9% vs 78.1%) *and* 2.8× larger (1.57M vs 565k). The classic cut-based top CR is the **worst of the four**, and carries 35× the signal contamination of the MVA CR.

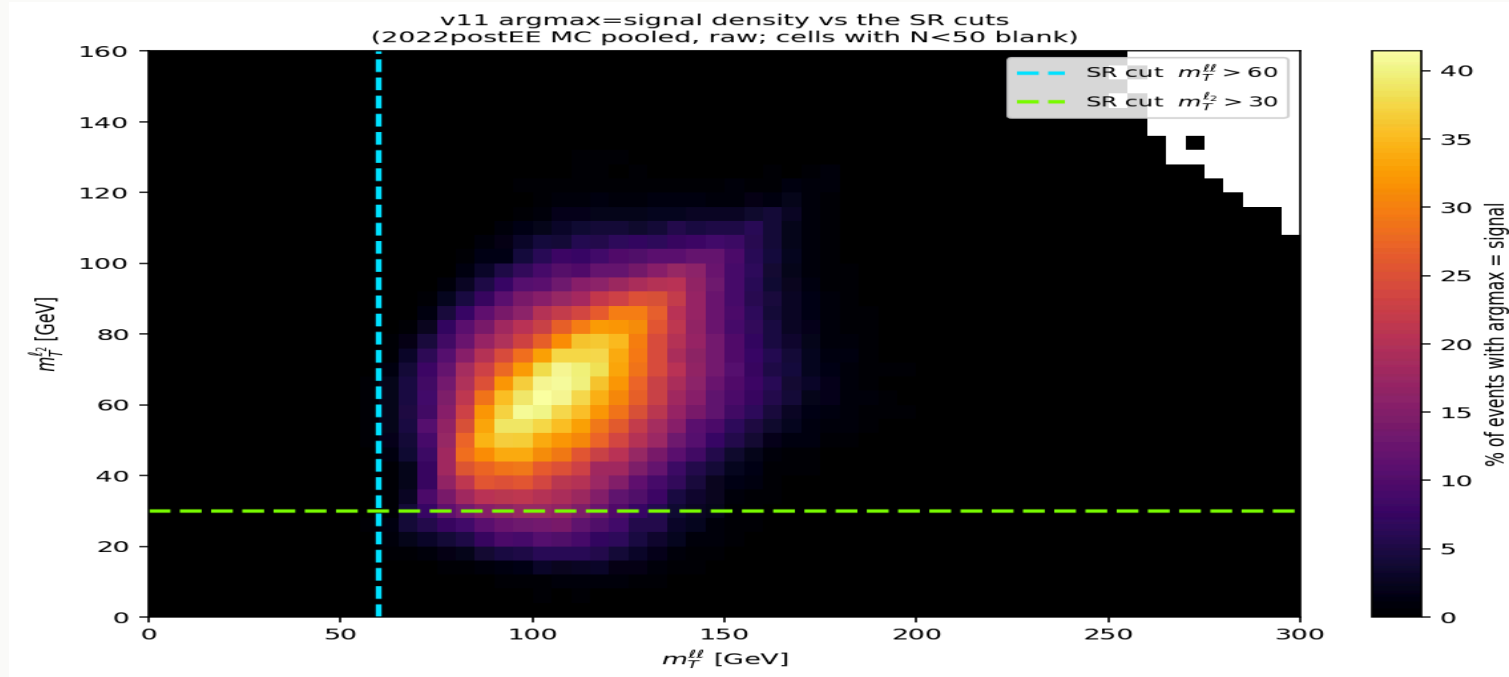
01b • What is actually in the MVA $t\bar{t}$ CR

TRUE PROCESS	N	FRACTION
$t\bar{t}$	1,375,464	87.86%
single top	146,891	9.38%
Higgs bkg	41,122	2.63%
diboson	1,872	0.12%
V+jets	105	0.01%
H+c (signal)	7	0.00%

Single-top is the only real contaminant — and it is physically $t\bar{t}$ -like, which is exactly what you want a $t\bar{t}$ CR to constrain. Signal leakage is **7 events out of 1.57M**.

02 · The MVA has already internalised the SR kinematic cuts

Fraction of events called signal across the (m_{ll}, m_{Tll}) plane — no kinematic cut applied



BELOW THE m_{Tll} WALL

0.0098%

75 events of 662,103

ABOVE THE WALL

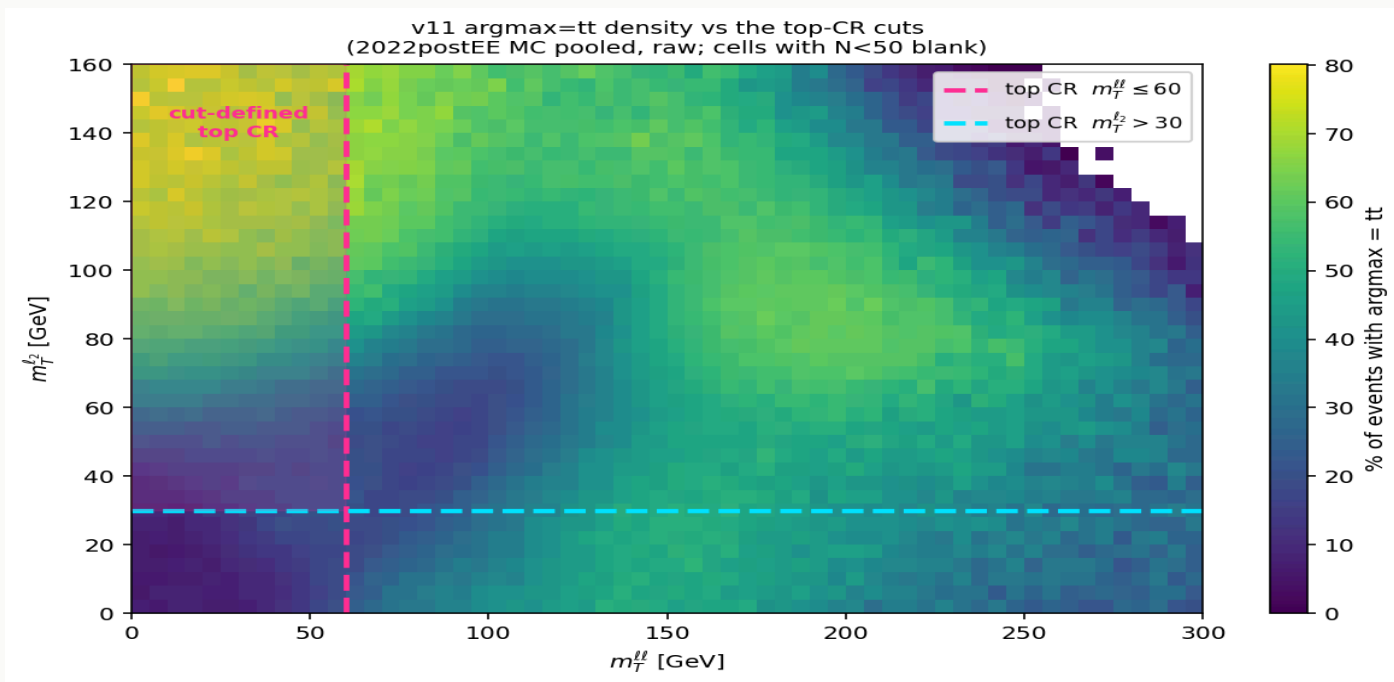
11.36%

1,160× higher

argmax=signal support is bounded at $m_{Tll} \in [52.8, 202]$ and $m_{ll} \leq 100$ —
despite `mtll` **not being an input feature**.

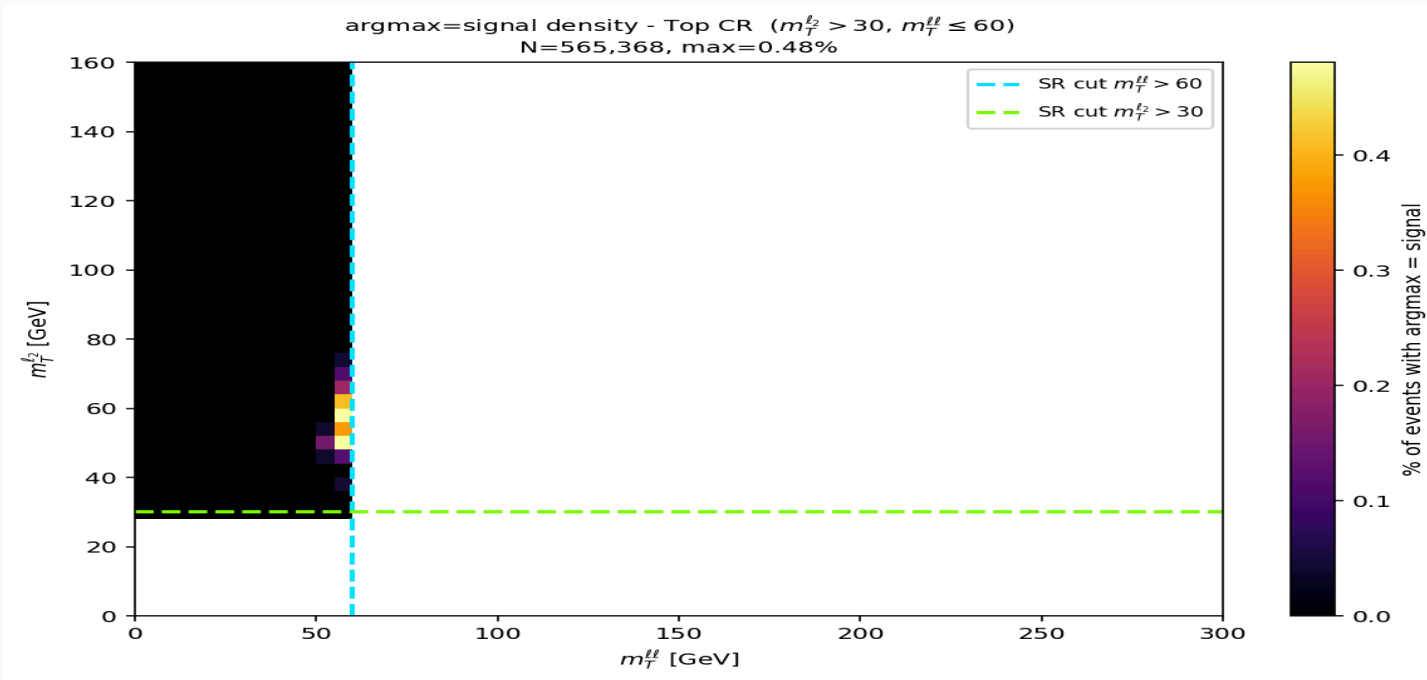
03 · Where the network puts $t\bar{t}$ — and where the hand-made CR sits

argmax= $t\bar{t}$ density across the same plane, with the cut-defined top CR overlaid



The $t\bar{t}$ class fills a **broad** region — the cut box ($m_{Tl}^{l1} \leq 60$ & $m_{Tl}^{l2} > 30$) sits inside it but captures only **15.3%** of all argmax= $t\bar{t}$ events. Purity inside the box is 42.5% vs 38.1% outside: the cut buys **+4 points of $t\bar{t}$ fraction while discarding 85% of the $t\bar{t}$ the network already identifies**. That is the whole case for the MVA-defined CR on one plot.

03b · Same behaviour in the top CR — separation is learned, not cut



Across the entire cut-defined top CR the argmax=signal fraction peaks at **0.48%**; only **66 of 565,368** events are called signal (0.0117%), max $P(H+c) = 0.358$ — below the 0.5 needed to ever win argmax.

The cut is redundant with the classifier. The network independently recovers the same region because both follow the same physics — not because it was trained on the cut.

`hww_MVA.yaml` has no m_T/m_{ll} cut in its base selection; labels are process truth.

04 · Cutting shrinks the training set the MVA needs

Efficiency of each kinematic cut, on top of the ≥ 1 c-jet requirement

CUT	E_S	E_B	$S/\sqrt{B}$	GAIN
$m_{T_{ll}} > 60$	0.926	0.790	0.00091	1.04×
$m_{T_{l2}} > 30$	0.890	0.827	0.00086	0.98×
$m_{ll} \leq 72$	0.978	0.397	0.00136	1.55×
all three (the SR)	0.845	0.271	0.00143	1.63×

The cost, made concrete. Putting $m_{ll} \leq 72$ into the base selection empties the high- m_{ll} CR *by construction* — a region holding **2.38M events at 83.0% $t\bar{t}$ purity**. You trade a well-populated constraint region for a 1.63× pre-selection number the MVA does not need.

05 · The real lever: the charm tag, not the kinematics

C-JET EFF — H+C SIGNAL

23.1%

medium WP discards 77% of signal

C-JET EFF — GGH

15.9%

the shape-degenerate competitor

ENRICHMENT H+C / GGH

1.46×

ggH:H+c 681:1 → 467:1

Loose vs medium, measured offline on the untagged tree (5,619 H+c events, `hww_2dcat_nocjet`)

C-TAG OPTION	SIGNAL N	EFF OF ≥1-JET BASE	VS MEDIUM
medium (CvL>0.160, CvB>0.304)	3,244	57.7%	1.00×
loose (CvL>0.054, CvB>0.182)	5,337	95.0%	1.65×
no tag at all	5,619	100.0%	1.73×

Loosening the WP recovers 1.65× the signal — most of what dropping the tag entirely would give (1.73×), while still rejecting light-flavour. Compare the kinematic cuts, worth 1.63× on S/√B but paid for with the control regions.

05b · The missing half: what the looser WP admits — and the answer is *no*

hww_ctag_compare, 2022preEE, full MC — four categories over **one** untagged jet collection, weighted yields

PROCESS	BASE (NO TAG)	MEDIUM WP	LOOSE WP	KIN, NO TAG
H+c (signal)	1.79×	1.00×	1.70×	1.56×
ggH (degenerate)	2.92×	1.00×	2.67×	2.16×
tt	1.44×	1.00×	1.40×	0.46×
V+jets	2.37×	1.00×	2.21×	0.38×

SIGNAL GAIN, MEDIUM → LOOSE

1.70×

what slide 05 measured

GGH GAIN, MEDIUM → LOOSE

2.67×

the half that was missing

H+C/GGH ENRICHMENT RETAINED

0.64×

loosening destroys 36% of it

ggH grows 1.57× faster than signal. The charm tag exists to buy H+c-over-ggH enrichment, and loosening the WP gives back **36%** of it. Dropping the tag entirely is worse (0.62×). The acceptance gain on slide 05 is real but it is **not free** — this is the measurement that decides it, and it says **keep the medium WP**.

06 · What this argument does not yet prove

- ~~Loosening the WP admits more ggH too.~~ **Now measured — see slide 05b.** ggH rises 2.67× against signal's 1.70×, so the enrichment falls to 0.64×. CvL carries the H+c-vs-ggH separation (AUC 0.731), CvB does not ($0.551 \approx$ coin flip); a looser WP moves down the CvL axis and admits the degenerate background faster than the signal.
- **Purity is measured here; sensitivity is not.** These are event counts, not a limit. The limit also depends on how systematics act on a larger background.
- **Selection and SF boundaries do not coincide.** The 2D SF scheme bins on $x = \text{CvL} / (\text{CvL} + \text{CvB}(1 - \text{CvL}))$ and $y = 1 - \text{CvB}$; a rectangular WP cut is a different shape in that plane. Already true in production, but worth stating.

Slides 01–05 are 2022postEE (hww_combine_fixed MVA tree, raw counts); **slide 05b is 2022preEE** (hww_ctag_compare , weighted yields, full MC). The two agree on the signal ratio to within a few percent across different eras, weighting, and code paths — the raw-postEE loose/medium of 1.65× against the weighted-preEE 1.70×.

Summary

1. **MVA-defined CRs are purer than cut-defined ones** — 87.9% vs 78.1% on $t\bar{t}$, and 2.8× the events.
2. **The MVA already applies the SR kinematic cuts internally** — 0.0098% vs 11.36% $\text{argmax}=\text{signal}$ across the $m_{T_{ll}}$ wall, with `mtll` not even an input.
3. **Cutting deletes the sidebands the network learns from**, and empties the CRs you need to fit.
4. **Neither acceptance lever is free.** Medium → loose recovers **1.70×** the signal — comparable to the 1.63× the kinematic cuts buy — but admits ggH at **2.67×**, giving back 36% of the H+c/ggH enrichment. **Keep the medium WP.** The kinematic cuts buy 1.63× on S/ $\sqrt{B}$ but empty the CRs. The MVA-defined regions remain the one change that costs nothing.

<footer>Slides 01–05: 2022postEE, v11 6-class MVA [hplusc, higgsbkg, tt, st, diboson, vjets], pooled MC 4,046,127 events, raw · Slide 05b: 2022preEE hww_ctag_compare, weighted yields, full MC · purity = true-process fraction</footer>